



May 30, 2025

Cindy Tomaszewski
City of Cadillac
1121 Plett Rd
Cadillac, MI 49601

RE: Project: Effluent PFAS
Pace Project No.: 50400684

Dear Cindy Tomaszewski:

Enclosed are the analytical results for sample(s) received by the laboratory on May 07, 2025. The results relate only to the samples included in this report. Results reported herein conform to the applicable TNI/NELAC Standards and the laboratory's Quality Manual, where applicable, unless otherwise noted in the body of the report.

The test results provided in this final report were generated by each of the following laboratories within the Pace Network:

- Pace Analytical Services - Minneapolis

If you have any questions concerning this report, please feel free to contact me.

Sincerely,

A handwritten signature in blue ink that reads "Brian Hall".

Brian Hall
brian.hall@pacelabs.com
(616)975-4500
Project Manager

Enclosures



REPORT OF LABORATORY ANALYSIS

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CERTIFICATIONS

Project: Effluent PFAS

Pace Project No.: 50400684

Pace Analytical Services, LLC - Minneapolis MN

1700 Elm Street SE, Minneapolis, MN 55414

Alabama Certification #: 40770

Alaska Contaminated Sites Certification #: 17-009

Alaska DW Certification #: MN00064

Arizona Certification #: AZ0014

Arkansas DW Certification #: MN00064

Arkansas WW Certification #: 88-0680

California Certification #: 2929

Colorado Certification #: MN00064

Connecticut Certification #: PH-0256

DoD Certification via A2LA #: 2926.01

EPA Region 8 Tribal Water Systems+Wyoming DW
Certification #: via MN 027-053-137

Florida Certification #: E87605

Georgia Certification #: 959

Idaho Certification #: MN00064

Illinois Certification #: 200011

Indiana Certification #: C-MN-01

Iowa Certification #: 368

ISO/IEC 17025 Certification via A2LA #: 2926.01

Kansas Certification #: E-10167

Kentucky DW Certification #: 90062

Kentucky WW Certification #: 90062

Louisiana DEQ Certification #: AI-03086

Louisiana DW Certification #: MN00064

Maine Certification #: MN00064

Maryland Certification #: 322

Michigan Certification #: 9909

Minnesota Certification #: 027-053-137

Minnesota Dept of Ag Approval: via MN 027-053-137

Minnesota Petrofund Registration #: 1240

Mississippi Certification #: MN00064

Missouri Certification #: 10100

Montana Certification #: CERT0092

Nebraska Certification #: NE-OS-18-06

Nevada Certification #: MN00064

New Hampshire Certification #: 2081

New Jersey Certification #: MN002

New York Certification #: 11647

North Carolina DW Certification #: 27700

North Carolina WW Certification #: 530

North Dakota Certification via A2LA #: R-036

North Dakota Certification via MN #: R-036

Ohio DW Certification #: 41244

Ohio VAP Certification (1700) #: CL101

Oklahoma Certification #: 9507

Oregon Primary Certification #: MN300001

Oregon Secondary Certification #: MN200001

Pennsylvania Certification #: 68-00563

Puerto Rico Certification #: MN00064

South Carolina Certification #: 74003001

Tennessee Certification #: TN02818

Texas Certification #: T104704192

Utah Certification #: MN00064

Vermont Certification #: VT-027053137

Virginia Certification #: 460163

Washington Certification #: C486

West Virginia DEP Certification #: 382

West Virginia DW Certification #: 9952 C

Wisconsin Certification #: 999407970

Wyoming UST Certification via A2LA #: 2926.01

USDA Permit #: P330-19-00208

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SAMPLE SUMMARY

Project: Effluent PFAS

Pace Project No.: 50400684

Lab ID	Sample ID	Matrix	Date Collected	Date Received
50400684001	Effluent	Water	05/06/25 11:10	05/07/25 10:10

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SAMPLE ANALYTE COUNT

Project: Effluent PFAS

Pace Project No.: 50400684

Lab ID	Sample ID	Method	Analysts	Analytes Reported	Laboratory
50400684001	Effluent	EPA 1633	AJG	65	PASI-M

PASI-M = Pace Analytical Services - Minneapolis

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SUMMARY OF DETECTION

Project: Effluent PFAS

Pace Project No.: 50400684

Lab Sample ID	Client Sample ID					
Method	Parameters	Result	Units	Report Limit	Analyzed	Qualifiers
50400684001	Effluent					
EPA 1633	PFBS	10.7	ng/L	3.1	05/28/25 18:32	
EPA 1633	PFHxA	6.3	ng/L	3.1	05/28/25 18:32	
EPA 1633	PFHxS	0.93J	ng/L	3.1	05/28/25 18:32	
EPA 1633	PFOS	1.3J	ng/L	3.1	05/28/25 18:32	
EPA 1633	PFOA	2.2J	ng/L	3.1	05/28/25 18:32	

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50400684

Sample: Effluent Lab ID: 50400684001 Collected: 05/06/25 11:10 Received: 05/07/25 10:10 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
EPA 1633F Water									
Analytical Method: EPA 1633 Preparation Method: EPA 1633									
Pace Analytical Services - Minneapolis									
11CI-PF3OUdS	<3.7	ng/L	12.3	3.7	1	05/24/25 09:05	05/28/25 18:32	763051-92-9	
3:3 FTCA	<4.2	ng/L	15.4	4.2	1	05/24/25 09:05	05/28/25 18:32	356-02-5	
4:2 FTS	<3.1	ng/L	12.3	3.1	1	05/24/25 09:05	05/28/25 18:32	757124-72-4	
5:3 FTCA	<18.4	ng/L	76.8	18.4	1	05/24/25 09:05	05/28/25 18:32	914637-49-3	
6:2 FTS	<3.6	ng/L	12.3	3.6	1	05/24/25 09:05	05/28/25 18:32	27619-97-2	
7:3 FTCA	<21.9	ng/L	76.8	21.9	1	05/24/25 09:05	05/28/25 18:32	812-70-4	
8:2 FTS	<3.2	ng/L	12.3	3.2	1	05/24/25 09:05	05/28/25 18:32	39108-34-4	
9CI-PF3ONS	<3.2	ng/L	12.3	3.2	1	05/24/25 09:05	05/28/25 18:32	756426-58-1	
ADONA	<3.6	ng/L	12.3	3.6	1	05/24/25 09:05	05/28/25 18:32	919005-14-4	
HFPO-DA	<3.4	ng/L	12.3	3.4	1	05/24/25 09:05	05/28/25 18:32	13252-13-6	
NEtFOSAA	<1.3	ng/L	3.1	1.3	1	05/24/25 09:05	05/28/25 18:32	2991-50-6	
NEtFOSA	<0.97	ng/L	3.1	0.97	1	05/24/25 09:05	05/28/25 18:32	4151-50-2	
NEtFOSE	<7.9	ng/L	30.7	7.9	1	05/24/25 09:05	05/28/25 18:32	1691-99-2	
NFDHA	<2.5	ng/L	6.1	2.5	1	05/24/25 09:05	05/28/25 18:32	151772-58-6	
NMeFOSAA	<0.96	ng/L	3.1	0.96	1	05/24/25 09:05	05/28/25 18:32	2355-31-9	
NMeFOSA	<0.95	ng/L	3.1	0.95	1	05/24/25 09:05	05/28/25 18:32	31506-32-8	
NMeFOSE	<7.9	ng/L	30.7	7.9	1	05/24/25 09:05	05/28/25 18:32	24448-09-7	
PFBS	10.7	ng/L	3.1	0.81	1	05/24/25 09:05	05/28/25 18:32	375-73-5	
PFDA	<0.95	ng/L	3.1	0.95	1	05/24/25 09:05	05/28/25 18:32	335-76-2	
PFHxA	6.3	ng/L	3.1	0.74	1	05/24/25 09:05	05/28/25 18:32	307-24-4	
PFBA	<3.0	ng/L	12.3	3.0	1	05/24/25 09:05	05/28/25 18:32	375-22-4	
PFDS	<1.3	ng/L	3.1	1.3	1	05/24/25 09:05	05/28/25 18:32	335-77-3	
PFDoS	<0.66	ng/L	3.1	0.66	1	05/24/25 09:05	05/28/25 18:32	79780-39-5	
PFEESA	<1.3	ng/L	6.1	1.3	1	05/24/25 09:05	05/28/25 18:32	113507-82-7	
PFHpS	<0.89	ng/L	3.1	0.89	1	05/24/25 09:05	05/28/25 18:32	375-92-8	
PFMBA	<1.5	ng/L	6.1	1.5	1	05/24/25 09:05	05/28/25 18:32	863090-89-5	
PFMPA	<1.2	ng/L	6.1	1.2	1	05/24/25 09:05	05/28/25 18:32	377-73-1	
PFNS	<0.91	ng/L	3.1	0.91	1	05/24/25 09:05	05/28/25 18:32	68259-12-1	
PFOSA	<0.94	ng/L	3.1	0.94	1	05/24/25 09:05	05/28/25 18:32	754-91-6	
PFPeA	<1.8	ng/L	6.1	1.8	1	05/24/25 09:05	05/28/25 18:32	2706-90-3	
PFPeS	<0.73	ng/L	3.1	0.73	1	05/24/25 09:05	05/28/25 18:32	2706-91-4	
PFDoA	<0.94	ng/L	3.1	0.94	1	05/24/25 09:05	05/28/25 18:32	307-55-1	
PFHpA	<1.2	ng/L	3.1	1.2	1	05/24/25 09:05	05/28/25 18:32	375-85-9	
PFHxS	0.93J	ng/L	3.1	0.79	1	05/24/25 09:05	05/28/25 18:32	355-46-4	
PFNA	<1.0	ng/L	3.1	1.0	1	05/24/25 09:05	05/28/25 18:32	375-95-1	
PFOS	1.3J	ng/L	3.1	0.75	1	05/24/25 09:05	05/28/25 18:32	1763-23-1	
PFOA	2.2J	ng/L	3.1	0.93	1	05/24/25 09:05	05/28/25 18:32	335-67-1	
PFTeDA	<0.79	ng/L	3.1	0.79	1	05/24/25 09:05	05/28/25 18:32	376-06-7	
PFTTrDA	<0.87	ng/L	3.1	0.87	1	05/24/25 09:05	05/28/25 18:32	72629-94-8	
PFUnA	<0.68	ng/L	3.1	0.68	1	05/24/25 09:05	05/28/25 18:32	2058-94-8	
Surrogates									
13C2-PFDoA (S)	54	%	10-130		1	05/24/25 09:05	05/28/25 18:32		
13C3HFPO-DA (S)	90	%	40-130		1	05/24/25 09:05	05/28/25 18:32		
13C3-PFBS (S)	89	%	40-135		1	05/24/25 09:05	05/28/25 18:32		
13C3-PFHxS (S)	81	%	40-130		1	05/24/25 09:05	05/28/25 18:32		

REPORT OF LABORATORY ANALYSIS

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ANALYTICAL RESULTS

Project: Effluent PFAS

Pace Project No.: 50400684

Sample: Effluent **Lab ID: 50400684001** Collected: 05/06/25 11:10 Received: 05/07/25 10:10 Matrix: Water

Parameters	Results	Units	PQL	MDL	DF	Prepared	Analyzed	CAS No.	Qual
EPA 1633F Water									
Analytical Method: EPA 1633 Preparation Method: EPA 1633									
Pace Analytical Services - Minneapolis									
Surrogates									
13C4-PFBA (S)	87	%.	5-130		1	05/24/25 09:05	05/28/25 18:32		
13C4-PFHpA (S)	84	%.	40-130		1	05/24/25 09:05	05/28/25 18:32		
13C5-PFHxA (S)	85	%.	40-130		1	05/24/25 09:05	05/28/25 18:32		
13C5-PFPeA (S)	90	%.	40-130		1	05/24/25 09:05	05/28/25 18:32		
13C6-PFDA (S)	77	%.	40-130		1	05/24/25 09:05	05/28/25 18:32		
13C8-PFOA (S)	80	%.	40-130		1	05/24/25 09:05	05/28/25 18:32		
13C8-PFOS (S)	78	%.	40-130		1	05/24/25 09:05	05/28/25 18:32		
13C8-PFOSA (S)	94	%.	40-130		1	05/24/25 09:05	05/28/25 18:32		
13C9-PFNA (S)	82	%.	40-130		1	05/24/25 09:05	05/28/25 18:32		
d3-MeFOSAA (S)	73	%.	40-170		1	05/24/25 09:05	05/28/25 18:32		
d3-NMeFOSA (S)	69	%.	10-130		1	05/24/25 09:05	05/28/25 18:32		
d5-EtFOSAA (S)	71	%.	25-135		1	05/24/25 09:05	05/28/25 18:32		
d5-NEtFOSA (S)	58	%.	10-130		1	05/24/25 09:05	05/28/25 18:32		
d7-NMeFOSE (S)	93	%.	10-130		1	05/24/25 09:05	05/28/25 18:32		
d9-NEtFOSE (S)	79	%.	10-130		1	05/24/25 09:05	05/28/25 18:32		
13C2-PFTA (S)	40	%.	10-130		1	05/24/25 09:05	05/28/25 18:32		
13C7-PFUdA (S)	68	%.	30-130		1	05/24/25 09:05	05/28/25 18:32		
13C24:2FTS (S)	72	%.	40-200		1	05/24/25 09:05	05/28/25 18:32		
13C26:2FTS (S)	83	%.	40-200		1	05/24/25 09:05	05/28/25 18:32		
13C28:2FTS (S)	74	%.	40-300		1	05/24/25 09:05	05/28/25 18:32		
13C3-PFPrA (S)	56	%.	5-130		1	05/24/25 09:05	05/28/25 18:32		

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50400684

QC Batch: 1008587

Analysis Method: EPA 1633

QC Batch Method: EPA 1633

Analysis Description: EPA 1633F Water

Laboratory: Pace Analytical Services - Minneapolis

Associated Lab Samples: 50400684001

METHOD BLANK: 5258138

Matrix: Water

Associated Lab Samples: 50400684001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
11CI-PF3OUdS	ng/L	<1.9	6.4	1.9	05/28/25 03:01	
3:3 FTCA	ng/L	<2.2	8.0	2.2	05/28/25 03:01	
4:2 FTS	ng/L	<1.6	6.4	1.6	05/28/25 03:01	
5:3 FTCA	ng/L	<9.6	40.0	9.6	05/28/25 03:01	
6:2 FTS	ng/L	<1.8	6.4	1.8	05/28/25 03:01	
7:3 FTCA	ng/L	<11.4	40.0	11.4	05/28/25 03:01	
8:2 FTS	ng/L	<1.7	6.4	1.7	05/28/25 03:01	
9CI-PF3ONS	ng/L	<1.7	6.4	1.7	05/28/25 03:01	
ADONA	ng/L	<1.9	6.4	1.9	05/28/25 03:01	
HFPO-DA	ng/L	<1.8	6.4	1.8	05/28/25 03:01	
NETFOSA	ng/L	<0.51	1.6	0.51	05/28/25 03:01	
NETFOSAA	ng/L	<0.66	1.6	0.66	05/28/25 03:01	
NETFOSE	ng/L	<4.1	16.0	4.1	05/28/25 03:01	
NFDHA	ng/L	<1.3	3.2	1.3	05/28/25 03:01	
NMeFOSA	ng/L	<0.49	1.6	0.49	05/28/25 03:01	
NMeFOSAA	ng/L	<0.50	1.6	0.50	05/28/25 03:01	
NMeFOSE	ng/L	<4.1	16.0	4.1	05/28/25 03:01	
PFBA	ng/L	<1.6	6.4	1.6	05/28/25 03:01	
PFBS	ng/L	<0.42	1.6	0.42	05/28/25 03:01	
PFDA	ng/L	<0.50	1.6	0.50	05/28/25 03:01	
PFDaA	ng/L	<0.49	1.6	0.49	05/28/25 03:01	
PFDoS	ng/L	<0.34	1.6	0.34	05/28/25 03:01	
PFDS	ng/L	<0.69	1.6	0.69	05/28/25 03:01	
PFEESA	ng/L	<0.69	3.2	0.69	05/28/25 03:01	
PFHpA	ng/L	<0.62	1.6	0.62	05/28/25 03:01	
PFHpS	ng/L	<0.47	1.6	0.47	05/28/25 03:01	
PFHxA	ng/L	<0.38	1.6	0.38	05/28/25 03:01	
PFHxS	ng/L	<0.41	1.6	0.41	05/28/25 03:01	
PFMBA	ng/L	<0.78	3.2	0.78	05/28/25 03:01	
PFMPA	ng/L	<0.62	3.2	0.62	05/28/25 03:01	
PFNA	ng/L	<0.52	1.6	0.52	05/28/25 03:01	
PFNS	ng/L	<0.47	1.6	0.47	05/28/25 03:01	
PFOA	ng/L	<0.49	1.6	0.49	05/28/25 03:01	
PFOS	ng/L	<0.39	1.6	0.39	05/28/25 03:01	
PFOSA	ng/L	<0.49	1.6	0.49	05/28/25 03:01	
PFPeA	ng/L	<0.95	3.2	0.95	05/28/25 03:01	
PFPeS	ng/L	<0.38	1.6	0.38	05/28/25 03:01	
PFTeDA	ng/L	<0.41	1.6	0.41	05/28/25 03:01	
PFTrDA	ng/L	<0.45	1.6	0.45	05/28/25 03:01	
PFUnA	ng/L	<0.36	1.6	0.36	05/28/25 03:01	

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REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50400684

METHOD BLANK: 5258138

Matrix: Water

Associated Lab Samples: 50400684001

Parameter	Units	Blank Result	Reporting Limit	MDL	Analyzed	Qualifiers
13C2-PFD _o A (S)	%	48	10-130		05/28/25 03:01	
13C2-PFTA (S)	%	49	10-130		05/28/25 03:01	
13C24:2FTS (S)	%	64	40-200		05/28/25 03:01	
13C26:2FTS (S)	%	62	40-200		05/28/25 03:01	
13C28:2FTS (S)	%	56	40-300		05/28/25 03:01	
13C3-PFBS (S)	%	48	40-135		05/28/25 03:01	
13C3-PFH _x S (S)	%	49	40-130		05/28/25 03:01	
13C3-PFPrA (S)	%	45	5-130		05/28/25 03:01	
13C3HFPO-DA (S)	%	51	40-130		05/28/25 03:01	
13C4-PFBA (S)	%	55	5-130		05/28/25 03:01	
13C4-PFH _p A (S)	%	48	40-130		05/28/25 03:01	
13C5-PFH _x A (S)	%	50	40-130		05/28/25 03:01	
13C5-PFP _e A (S)	%	52	40-130		05/28/25 03:01	
13C6-PFDA (S)	%	48	40-130		05/28/25 03:01	
13C7-PFUD _A (S)	%	50	30-130		05/28/25 03:01	
13C8-PFOA (S)	%	50	40-130		05/28/25 03:01	
13C8-PFOS (S)	%	47	40-130		05/28/25 03:01	
13C8-PFOSA (S)	%	50	40-130		05/28/25 03:01	
13C9-PFNA (S)	%	49	40-130		05/28/25 03:01	
d3-MeFOSAA (S)	%	50	40-170		05/28/25 03:01	
d3-NMeFOSA (S)	%	36	10-130		05/28/25 03:01	
d5-EtFOSAA (S)	%	50	25-135		05/28/25 03:01	
d5-NEtFOSA (S)	%	37	10-130		05/28/25 03:01	
d7-NMeFOSE (S)	%	43	10-130		05/28/25 03:01	
d9-NEtFOSE (S)	%	43	10-130		05/28/25 03:01	

LABORATORY CONTROL SAMPLE: 5258139

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
11CI-PF3OUdS	ng/L	75.2	73.9	98	55-160	
3:3 FTCA	ng/L	99.2	92.7	93	65-130	
4:2 FTS	ng/L	75.2	75.7	101	70-145	
5:3 FTCA	ng/L	496	485	98	70-135	
6:2 FTS	ng/L	76.8	78.7	102	65-155	
7:3 FTCA	ng/L	496	468	94	50-145	
8:2 FTS	ng/L	76.8	80.9	105	60-150	
9CI-PF3ONS	ng/L	75.2	78.3	104	70-155	
ADONA	ng/L	75.2	76.1	101	65-145	
HFPO-DA	ng/L	80	80.5	101	70-140	
NEtFOSA	ng/L	19.2	18.1	95	65-145	
NEtFOSAA	ng/L	19.2	18.5	97	70-145	
NEtFOSE	ng/L	192	190	99	70-135	
NFDHA	ng/L	40	39.6	99	50-150	
NMeFOSA	ng/L	19.2	18.9	99	60-150	

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QUALITY CONTROL DATA

Project: Effluent PFAS

Pace Project No.: 50400684

LABORATORY CONTROL SAMPLE: 5258139

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
NMeFOSAA	ng/L	19.2	17.9	93	50-140	
NMeFOSE	ng/L	192	189	98	70-145	
PFBA	ng/L	80	77.3	97	70-140	
PFBS	ng/L	17.6	17.2	98	60-145	
PFDA	ng/L	19.2	19.5	102	70-140	
PFDaA	ng/L	19.2	19.7	103	70-140	
PFDoS	ng/L	19.2	20.3	106	50-145	
PFDS	ng/L	19.2	17.4	91	60-145	
PFEESA	ng/L	35.2	34.1	97	70-140	
PFHpA	ng/L	19.2	18.7	97	70-150	
PFHpS	ng/L	19.2	18.4	96	70-150	
PFHxA	ng/L	19.2	18.6	97	70-145	
PFHxS	ng/L	17.6	17.1	97	65-145	
PFMBA	ng/L	40	38.9	97	60-150	
PFMPA	ng/L	40	38.0	95	55-140	
PFNA	ng/L	19.2	18.4	96	70-150	
PFNS	ng/L	19.2	18.2	95	65-145	
PFOA	ng/L	19.2	19.3	100	70-150	
PFOS	ng/L	19.2	16.5	86	55-150	
PFOSA	ng/L	19.2	19.4	101	70-145	
PFPeA	ng/L	40	38.7	97	65-135	
PFPeS	ng/L	19.2	17.4	90	65-140	
PFTeDA	ng/L	19.2	19.7	103	60-140	
PFTrDA	ng/L	19.2	18.9	98	65-140	
PFUnA	ng/L	19.2	18.8	98	70-145	
13C2-PFDaA (S)	%			65	10-130	
13C2-PFTA (S)	%			66	10-130	
13C24:2FTS (S)	%			84	40-200	
13C26:2FTS (S)	%			82	40-200	
13C28:2FTS (S)	%			75	40-300	
13C3-PFBS (S)	%			68	40-135	
13C3-PFHxS (S)	%			70	40-130	
13C3-PFPrA (S)	%			76	5-130	
13C3HFPO-DA (S)	%			70	40-130	
13C4-PFBA (S)	%			75	5-130	
13C4-PFHpA (S)	%			73	40-130	
13C5-PFHxA (S)	%			72	40-130	
13C5-PFPeA (S)	%			73	40-130	
13C6-PFDA (S)	%			67	40-130	
13C7-PFUdA (S)	%			68	30-130	
13C8-PFOA (S)	%			71	40-130	
13C8-PFOS (S)	%			67	40-130	
13C8-PFOSA (S)	%			69	40-130	
13C9-PFNA (S)	%			70	40-130	
d3-MeFOSAA (S)	%			70	40-170	
d3-NMeFOSA (S)	%			53	10-130	
d5-EtFOSAA (S)	%			70	25-135	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA

Project: Effluent PFAS
Pace Project No.: 50400684

LABORATORY CONTROL SAMPLE: 5258139						
Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limits	Qualifiers
d5-NEtFOSA (S)	%.			54	10-130	
d7-NMeFOSE (S)	%.			66	10-130	
d9-NEtFOSE (S)	%.			64	10-130	

Results presented on this page are in the units indicated by the "Units" column except where an alternate unit is presented to the right of the result.

REPORT OF LABORATORY ANALYSIS



QUALIFIERS

Project: Effluent PFAS
Pace Project No.: 50400684

DEFINITIONS

DF - Dilution Factor, if reported, represents the factor applied to the reported data due to dilution of the sample aliquot.

ND - Not Detected at or above adjusted reporting limit.

TNTC - Too Numerous To Count

J - Estimated concentration above the adjusted method detection limit and below the adjusted reporting limit.

MDL - Adjusted Method Detection Limit.

PQL - Practical Quantitation Limit.

RL - Reporting Limit - The lowest concentration value that meets project requirements for quantitative data with known precision and bias for a specific analyte in a specific matrix.

S - Surrogate

1,2-Diphenylhydrazine decomposes to and cannot be separated from Azobenzene using Method 8270. The result for each analyte is a combined concentration.

Consistent with EPA guidelines, unrounded data are displayed and have been used to calculate % recovery and RPD values.

LCS(D) - Laboratory Control Sample (Duplicate)

MS(D) - Matrix Spike (Duplicate)

DUP - Sample Duplicate

RPD - Relative Percent Difference

NC - Not Calculable.

SG - Silica Gel - Clean-Up

U - Indicates the compound was analyzed for, but not detected.

N-Nitrosodiphenylamine decomposes and cannot be separated from Diphenylamine using Method 8270. The result reported for each analyte is a combined concentration.

Reported results are not rounded until the final step prior to reporting. Therefore, calculated parameters that are typically reported as "Total" may vary slightly from the sum of the reported component parameters.

Pace Analytical is TNI accredited. Contact your Pace PM for the current list of accredited analytes.

TNI - The NELAC Institute.

REPORT OF LABORATORY ANALYSIS

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QUALITY CONTROL DATA CROSS REFERENCE TABLE

Project: Effluent PFAS
Pace Project No.: 50400684

Lab ID	Sample ID	QC Batch Method	QC Batch	Analytical Method	Analytical Batch
50400684001	Effluent	EPA 1633	1008587	EPA 1633	1009387

REPORT OF LABORATORY ANALYSIS

CHAIN-OF-CUSTODY Analytical Request Document
Chain-of-Custody is a LEGAL DOCUMENT - Complete all relevant fields

Company Name:	City of Cadillac
Street Address:	1121 Plett Rd Cadillac, MI 49601
Customer Project #:	
Project Name:	Effluent PFAS
Site Collection Info/Facility ID (as applicable):	
Time Zone Collected:	[] AK [] PT [] MT [] CT [] ET
Data Deliverables:	Regulatory Program (DW, RCRA, etc.) as applicable: Reportable [] Yes [] No
[] Level II [] Level III [] Level IV	
[] EQUIS	
[] Other	

Customer Sample ID	Matrix *	Comp / Grab	Composite Start	Collected or Composite End	# Cont.	Res. Chlorine
EFFluent	ww	4	5-6-25 11:10		3	

Date Results Requested: [] Same Day [] 1 Day [] 2 Day [] 3 Day [] Other [] Yes [] No

Field Filtered (if applicable): [] Yes [] No

Analysis:

Regulatory Program (DW, RCRA, etc.) as applicable: Reportable [] Yes [] No

County / State origin of sample(s): Michigan

Invoice To: Cindy Tomaszewski

Invoice E-Mail: ctomaszewski@cadillac-mi.net

Purchase Order # (if applicable):

Quote #:

Container Size: (1) 1L, (2) 500mL, (3) 250mL, (4) 125mL, (5) 100mL, (6) 40mL vial, (7) Encore, (8) TerraCore, (9) 90mL, (10) Other

Preservative Types: (1) None, (2) HNO3, (3) H2SO4, (4) HCl, (5) NaOH, (6) Zn Acetate, (7) NaHSO4, (8) Sod. Thiosulfate, (9) Ascorbic Acid, (10) MeOH, (11) Other

Proj. Mgr: Brian Hall

AcctNum / Client ID:

Table #:

Profile / Template: 7675

Prelog / Bottle Ord. ID: EZ 3124086

Sample Comment

Grab.

1633 PFAS - Pace MN

X

Additional Instructions from Pace®:
1633 PFAS - 40 Compounds
Requires Reporting by 10th of Following Month

Collected By: Taysha Laczko

Signature: T. Laczko

Date/Time: 5-6-25 11:36

Received by Company: Signature

Date/Time: 5-6-25 11:36

Received by Company: Signature

Date/Time: 5-6-25 11:36

Relinquished by Company: Signature

Signature: T. Laczko

Date/Time: 5-6-25 11:36

Received by Company: Signature

Date/Time: 5-6-25 11:36

Received by Company: Signature

Date/Time: 5-6-25 11:36

Received by Company: Signature

Relinquished by Company: Signature

Signature: T. Laczko

Date/Time: 5-6-25 11:36

Received by Company: Signature

Date/Time: 5-6-25 11:36

Received by Company: Signature

Date/Time: 5-6-25 11:36

Received by Company: Signature

Pace® Location Requested (City/State):
Pace Analytical Grand Rapids
4171 40th Street SE, Grand Rapids, MI 49512

WU#: 50400684

50400684

Specify Container Size **

Identify Container Preservative Type**

Analysis Requested

Proj. Mgr: Brian Hall

AcctNum / Client ID:

Table #:

Profile / Template: 7675

Prelog / Bottle Ord. ID: EZ 3124086

Sample Comment

Grab.

1633 PFAS - Pace MN

X

Customer Remarks / Special Conditions / Possible Hazards:

Coolers:

Thermometer ID:

Correction Factor (°C):

Obs. Temp. (°C):

Corrected Temp. (°C):

On Ice:

Tracking Number:

Date/Time: 5/7/25 10:10

Delivered by: [] In-Person [] Courier

Date/Time: 5/7/25 10:10

Received by Company: Signature

Date/Time: 5/7/25 10:10

Received by Company: Signature

Date/Time: 5/7/25 10:10

Relinquished by Company: Signature

Signature: T. Laczko

Date/Time: 5/7/25 10:10

Received by Company: Signature

Date/Time: 5/7/25 10:10

Received by Company: Signature

Date/Time: 5/7/25 10:10

Received by Company: Signature



Sample Conditions Upon Receipt Form (SCUR)

Date/Time: 5/7 1233	Evaluated By: DAS	MO# : 50400684	
Client: City of Cadillac	PM: BJH	Due Date: 06/06/25	
Lab Notified of Rush or Short Holds: YES NO NA	CLIENT: GR-Cadillac		
Project Received Via: FedEx UPS Client Pace Courier Other: _____		Comments:	
Custody Seal Present and Intact:	YES	NO	NA
Received Sample Information Form (SIF): Drinking Waters Only	YES	NO	NA
Short Hold Present (≤ 48 Hours):	YES	NO	
Sample Received in Hold:	YES	NO	
Custody Signature Present:	YES	NO	
Collector Signature Present:	YES	NO	
Sample Collected Today and On Ice:	YES	NO	N/A
IR Gun #: 350 351 Therm #: 282 283	Temp. should be 0°C - 6°C (Initial/Corrected)		
Ice Type: WET Bagged / WET Loose BLUE NONE	1. Cooler Temp. Upon Receipt: 2.6/2.6 °C		
Ice Location: TOP BOTTOM MIDDLE DISPERSED	2. Cooler Temp. Upon Receipt: _____ °C		
Temp Blank Received:	YES	NO	
Sample Label Matches COC (ID/Date/Time):	YES	NO	
Container Intact:	YES	NO	
Correct Container:	YES	NO	
Sufficient Volume:	YES	NO	
Sample pH Acceptable: All containers needing preservation are found to be in compliance with EPA recommendation. Denote with red dot on container lid if unacceptable. pH Strip Lot #: _____ Exceptions are VOA, coliform, LLHg, O&G/TPH, or any container with a septum cap or preserved with HCl	YES	NO	N/A
Residual Chlorine Absent: Cl ₂ Strip Lot #: _____ Applies to SVOC 625, PCB/Pest. 608, Total/Amenable/Available/Free Cyanide	YES	NO	N/A
VOA Headspace Acceptable (<6mm): Denote with silver x on rim of container cap if unacceptable.	YES	NO	N/A
Trip Blank Received: HCl MeOH Other: _____	YES	NO	ON HOLD
Comments:	3. Cooler Temp. Upon Receipt: _____ °C		
	4. Cooler Temp. Upon Receipt: _____ °C		
	Non-Conformance Form Required: YES NO		